



2026-10-02 00:35 EDT · 3 samples · 3 ok, 0 with warnings, 0 failed · operator: Example

Sample	Spoligotype (octal)	SB / SIT	Species	Lineage	Pattern (spacers 1 to 43)	Status
AF2122_97	664073777777600	SB0140 SIT683	M. bovis	BOV		OK
NC_002945	664073777777600	SB0140 SIT683	M. bovis	BOV		OK
H37Rv	777777477760771	SIT451	M. tuberculosis	4.9		OK

Reviewed by		Date		Signature	
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Species and lineage

+ / -: DNA of the region present in / absent from the sample, from 100 bp segments inside the region (not from amplicon sizes). See "Definitions and methods".

Reads per spacer

Blue: present (count ≥ 5). Orange: called absent but seen in some reads.

Species and lineage

Region	H37Rv region	Usually deleted in	Segments found	Depth vs control	Result
RD1	4,350,251-4,359,740	BCG, Dassie bacillus	20 of 20	1.00	+ present
RD4	1,696,001-1,708,740	M. bovis, BCG (some M. canettii)	0 of 20	0.00	– deleted
RD7	2,208,001-2,220,740	M. africanum lineage 6, animal lineages	0 of 20	0.00	– deleted
RD9	2,330,051-2,332,140	M. africanum (lineages 5, 6), animal lineages	0 of 8	0.00	– deleted
RD12	3,485,101-3,487,540	M. bovis, BCG, M. caprae, M. orygis (some M. canettii)	0 of 20	0.00	– deleted

Lineage SNP	Position (H37Rv)	Gene	Reads with lineage allele	Other reads
BOV_AFRI	1,882,180	Rv1662	1	0
BOV	2,831,482	Rv2515c	1	0

Spacer	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22
Count	1	1	0	1	1	0	1	0	0	0	0	0	1	1	1	0	1	1	1	1	1	1
Spacer	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	
Count	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	

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H37Rv OK

Spoligotype (octal)	777777477760771
Hexadecimal	7F-7F-7C-7F-F0-7F
Binary	1111111111111111100111111111100001111111
Pattern	 1 5 10 15 20 25 30 35 40 43
SB number (Mbovis.org)	Not in Mbovis.org: the pattern has no name in this database
SIT (SITVIT2)	SIT451 · SITVIT2 family T-H37Rv
Data	assembly (fasta)
Input file	/example/data/H37Rv.fasta 4.3 MB · modified 2026-09-24 09:13:56 EDT · MD5 e51ec80710a07b144cc60affab8a208f
Input size	1 contig, 4,411,532 bases
Minimum count	1 contig per spacer to call it present
Present spacers	37 of 43, median count 1
Run time	2.0 s

Species and lineage

Species	M. tuberculosis
Lineage	4.9 · Euro-American (H37Rv-like) · typical spoligotypes: T1
MTBC DNA	median 1 contig per control region, 100% of the control regions found

Region	H37Rv region	Usually deleted in	Segments found	Depth vs control	Result
RD1	4,350,251-4,359,740	BCG, Dassie bacillus	20 of 20	1.00	+ present
RD4	1,696,001-1,708,740	M. bovis, BCG (some M. canettii)	20 of 20	1.00	+ present
RD7	2,208,001-2,220,740	M. africanum lineage 6, animal lineages	20 of 20	1.00	+ present
RD9	2,330,051-2,332,140	M. africanum (lineages 5, 6), animal lineages	8 of 8	1.00	+ present
RD12	3,485,101-3,487,540	M. bovis, BCG, M. caprae, M. orygis (some M. canettii)	20 of 20	1.00	+ present

+ / -: DNA of the region present in / absent from the sample, from 100 bp segments inside the region (not from amplicon sizes). See "Definitions and methods".

Lineage SNP	Position (H37Rv)	Gene	Reads with lineage allele	Other reads
4	931,123	Rv0835	1	0
4.9	1,759,252	Rv1552	1	0

Contigs per spacer

Spacer	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22
Count	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0	1
Spacer	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	
Count	1	1	1	1	1	1	1	1	1	1	0	0	0	0	1	1	1	1	1	1	1	

Blue: present (count ≥ 1). Orange: called absent but seen in some contigs.

Definitions and methods

Spoligotype	The presence (1, filled square) or absence (0, empty square) of the 43 spacers of the direct repeat (DR) locus, written as a 43-digit binary pattern, a 15-digit octal code (Dale et al. 2001) or a hexadecimal code. These codes are universal: they identify the pattern itself, and are the ones to use to exchange spoligotypes.
SB number	The name of a pattern in the Mbovis.org database of spoligotypes of the RD9-deleted lineages (M. bovis and other animal-adapted lineages). "Not in Mbovis.org" only means that this database has no name for the pattern, as for all M. tuberculosis patterns: it does not mean that the spoligotype is invalid or new.
SIT	Shared international type of the SITVIT2 database (Institut Pasteur de Guadeloupe), from the SITVIT2 patterns published with SpolLineages (9,656 patterns, 3,850 SITs, 2022 list). "Orphan": a SITVIT2 pattern without SIT. "Not in SITVIT2 list": the pattern is not in this list, which does not include the SITs created since 2022. SITVIT2 lacks many patterns of the animal-adapted lineages: only 603 of the 1,976 Mbovis.org patterns have a SIT in this list. For these lineages, the SB number is the reference name, and no closest SIT is given for a pattern with an SB number.
Spoligotype family	SITVIT2 family of the pattern (e.g. Beijing, LAM3, BOV_1), and the spoligotype families typical of the lineage (Coll et al. 2014). Families are labels of patterns, not phylogenetic lineages.
Lineage	From the 62 SNPs of the barcode of Coll et al. (2014): the reads carrying each allele are counted with exact 31-mers; a lineage is called when at least 80% of the reads (at least 3, or 1 contig) carry its allele.
Livestock lineage	Lineage of the livestock-associated MTBC after Zwyrer et al. (2021): La1 (M. bovis), La2 (M. caprae), La3 (M. orygis), and the La1 sublineages La1.1 to La1.8, with their former names (e.g. La1.8.1: clonal complex Eu1; BCG belongs to La1.2). From 88 of the SNPs marked as markers in the extended data of the paper (4 or 5 per group), counted as above; a group is called when at least 2 of its SNPs carry the derived allele, as in the KvarQ test suite of the paper. La2 and La3 tell M. caprae from M. orygis, which have the same RD profile.

Regions of difference

Regions of difference (RD) are determined in silico, from the sequencing data, without PCR. Each region is the part of the H37Rv genome missing from M. bovis AF2122/97 (RD4, RD7, RD9, RD12) or from BCG Pasteur (RD1), and is represented by 100 bp segments inside it (table below), chosen to be found in all the MTBC genomes that have the region and in none of those lacking it or of six non-tuberculous mycobacteria. A segment is found when Seal (25-mers, 1 mismatch) finds it in the reads at a depth of at least 5% of the depth of MTBC-specific control regions, or in the assembly. A region is present when all its segments are found (in reads, up to 10% of them, at least 1, may be missing: low depth); deleted when at most 10% of its segments (at least 1: stray reads) are found; partially deleted in between, with the H37Rv coordinates of the missing segments; present at reduced depth when, in reads, its segments are found at less than 50% of the control depth (mixed sample?).

Region	H37Rv region (NC_000962.3)	Segments	Usually deleted in
RD1	4,350,251-4,359,740 (9,490 bp)	20	BCG, Dassie bacillus
RD4	1,696,001-1,708,740 (12,740 bp)	20	M. bovis, BCG (some M. canettii)
RD7	2,208,001-2,220,740 (12,740 bp)	20	M. africanum lineage 6, animal lineages
RD9	2,330,051-2,332,140 (2,090 bp)	8	M. africanum (lineages 5, 6), animal lineages
RD12	3,485,101-3,487,540 (2,440 bp)	20	M. bovis, BCG, M. caprae, M. orygis (some M. canettii)

The + and – of the RD profile therefore mean that the DNA of the region is present or absent, like a PCR with primers inside the region (amplification = present). spoligotyper does not measure amplicon sizes or deletion junctions: assays that distinguish RDs by the size of an amplicon spanning the region may report partial or strain-specific deletions differently. A partially deleted region counts as + when at least half of its segments are found: for example, M. microti lacks the part of RD1 inside its own RD1mic deletion (reported as partially deleted) and is RD1 + in the RD profile, as in the classical RD PCR scheme. The species is read from the RD profile, refined with the lineage SNPs and the spacers.

Spacers and codes

Spacers are counted with Seal (BBTools): each of the 43 spacers is searched as a single 25-mer on both strands, allowing 1 mismatch (k=25, hdist=1, rcomp=t, maskmiddle=f, ambiguous=all). A spacer is present when it is found in at least the minimum count of reads (contigs for assemblies). Known variants of standard spacers, too different to be found as the spacer (more than 1 mismatch) but named as present in the spoligotype databases, are searched the same way and counted for their spacer (e.g. spacer 3 of M. orygis, 2 mismatches): the sample page says so. The fraction of MTBC reads is the depth of the control regions divided by the depth expected from the number of bases.

Run information

Operator	Example
User	example
Computer	example
Operating system	Linux-6.8.0-139-generic-x86_64-with-glibc2.39
Started	2026-10-02 00:35:41 EDT
Finished	2026-10-02 00:35:54 EDT
Duration	12.9 s
Working directory	/example
Command	spoligotyper -i data/ -o results/ -t 8 --operator Example

Parameters

Input	/example/data
Output folder	/example/results
Minimum count	default (5 for fastq, 1 for fasta)
Threads	8
Jobs	1
Seal memory	1g
Species and lineage	yes

Software

spoligotyper	0.8.0
Python	3.14.7 (/opt/conda/bin/python3.14)
Seal	/opt/conda/bin/seal.sh
BBTools	40.02
Java	openjdk version "25.0.2-internal" 2026-01-20

Reference data

Spoligotype database	/opt/conda/lib/python3.14/site-packages/spoligotyper/data/spoligotype_db.txt 1,976 patterns · MD5 0a88f7b89f746de2a11e1b4af0151e36
Spacer sequences	/opt/conda/lib/python3.14/site-packages/spoligotyper/data/spoligo_spacers.fasta 43 spacers · MD5 033c08e66aaa90193973f8d542e94c1
Spacer variants	/opt/conda/lib/python3.14/site-packages/spoligotyper/data/spacer_variants.fasta 1 known variants · MD5 2107b897bc759ce9d96a5158a9909f3f
Species markers	/opt/conda/lib/python3.14/site-packages/spoligotyper/data/markers.fasta MD5 87135cd99f26936911997867c6f1964b
Lineage SNP barcode	/opt/conda/lib/python3.14/site-packages/spoligotyper/data/lineage_barcode.tsv MD5 f676fb8f470889ae02cb2f2b6c5970d7
Lineage SNP sequences	/opt/conda/lib/python3.14/site-packages/spoligotyper/data/lineage_snps.fasta MD5 7e662a9baee450e79065705eb7b79021
Livestock lineage SNP barcode	/opt/conda/lib/python3.14/site-packages/spoligotyper/data/livestock_barcode.tsv MD5 042f0761e4ac34e498c99653f514772b
Livestock lineage SNP sequences	/opt/conda/lib/python3.14/site-packages/spoligotyper/data/livestock_snps.fasta MD5 8f8ea8ccbaacf13349f67fbe2cce5edb
SIT database	/opt/spoligotyper/sit_database.tsv 9,656 patterns, 3,850 SITs · SHA-256 9f3daf512900f2e68902c3ba25cd84080b0a94f9604775c79a3f61d6cfbe39b2 SIT database for spoligotyper, converted from SpolLineages Spoligo_list.csv (SITVIT2 patterns)

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